Beyond Feature Attribution: Do Deep Learning Models Learn Verifiable and Transferable Hydrological Knowledge?

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Abstract

In the context of the widespread application of deep learning in streamflow prediction, whether these models truly learn hydrologically meaningful information remains unclear. Existing studies largely rely on post hoc techniques such as SHAP and LIME, lacking systematic and verifiable interpretability analysis, and failing to assess whether model responses are physically consistent or whether the extracted knowledge is transferable across models.To address this gap, this study treats interpretability as a primary objective and proposes a Sparse Transformer-based Explainable Hydrology framework (STEH). The framework links inputs, internal circuit-level structures, and outputs through structural sparsification and pathway-level analysis, enabling interpretable characterization. It further incorporates circuit-level ablation, sensitivity analysis, and cross-model validation to evaluate the verifiability, behavioral consistency, and transferability of learned knowledge.Experiments across large-scale basins show that structurally simpler models exhibit clearer internal pathways and stronger physical interpretability, while more complex models may introduce redundancy and even degrade performance. Moreover, physically meaningful information is consistently identified across models with varying performance levels. The extracted structural knowledge can be transferred to improve traditional machine learning models.These findings suggest that deep learning can serve not only as a predictive tool but also as apotential source of hydrological insight, highlighting the need to move from descriptive explanation toward systematic verification in interpretability research.

Keywords: Model interpretability, Explainable artificial intelligence, Structural sparsity, Hydrological modeling, Streamflow prediction, Transformer

1 Introduction

Streamflow prediction plays a critical role in water resource management, flood control, and disaster mitigation, as well as basin operation and regulation[1]. Under intensifying hydro-climatic uncertainty[2], such predictions increasingly support high-stakes decision-making processes, including water allocation, flood-risk governance, and adaptive basin management[3]. In these contexts, model outputs not only guide engineering operations but may also influence regional or even transboundary water resource negotiations. As a result, model trustworthiness has become as important as predictive accuracy.

In recent years, machine learning and deep learning models have achieved significant improvements in streamflow prediction, demonstrating strong capability in capturing complex nonlinear relationships and temporal dependencies[4, 5]. However, these models are often criticized as "black boxes"[6], whose internal decision-making processes are difficult to interpret intuitively, thereby limiting the transparency and reliability of predictions[7]. In non-stationary or extrapolation scenarios, even highly accurate models may produce unreliable predictions if their internal logic is not understood, potentially misleading basin management decisions. Therefore, improving interpretability while maintaining predictive performance has become a central challenge in hydrological modeling[8].

To mitigate the black-box nature of data-driven models, researchers have increasingly incorporated hydrological process knowledge into neural networks to enhance physical consistency[9]. A representative approach is Physics-Informed Neural Networks (PINNs), which embed governing physical equations into the loss function as constraints, enabling models to satisfy both observational data and physical laws[10, 11]. For example, Fang et al.[12] proposed the SWAN model, a PINN-based framework that integrates saturated streamflow process knowledge into the neural network architecture. While such approaches improve physical plausibility to some extent, physical consistency does not inherently guarantee interpretability[13]. The internal mechanisms by which models utilize input variables[14] and represent hydrological processes remain largely unclear[15].

Current mainstream interpretability methods primarily rely on post-hoc analysis techniques, such as LIME[16] and SHAP[17]. These approaches interpret complex models by constructing local surrogate approximations or decomposing predictions based on cooperative game theory[18], thereby quantifying the contribution of input features to model outputs. In this way, they enhance the transparency of otherwise black-box models to a certain extent[19]. In streamflow prediction studies, such methods provide intuitive tools for analyzing the influence of key variables, including precipitation, evapotranspiration, and soil moisture, and have consequently become among the most widely adopted interpretability techniques.

However, the explanations produced by these methods generally lack explicit verifiability[20]. Unlike model predictions, which can be directly evaluated against observed streamflow, feature attributions do not have an objective ground truth and are therefore difficult to validate through independent experiments[21]. Existing studies have shown that even when a model achieves high predictive accuracy, its explanations may partly reflect complex statistical fitting or spurious correlations[22]. As a result, relying solely on feature attribution makes it difficult to determine whether a model has genuinely learned physically meaningful relationships[23], which limits its credibility in high-stakes decision-making contexts.

Furthermore, most existing studies remain at the level of variable-level analysis based on SHAP or LIME[24, 25], lacking a systematic evaluation of model behavior. While these methods provide useful insights into the relative importance of input variables, they do not assess how models respond under changing conditions[26]. In hydrological systems, however, understanding model behavior is essential. A physically reliable model is expected to produce directionally consistent and hydrologically reasonable responses[27] to variations in key drivers, such as precipitation.

Specifically, limited attention has been given to how models behave under input perturbations or counterfactual scenarios, resulting in a lack of systematic sensitivity analysis[28]. For example, when precipitation magnitude increases, streamflow volume should correspondingly increase in a stable and consistent manner[29]. If a model fails to exhibit such reasonable response patterns under varying input magnitudes, its explanations—even if statistically consistent—cannot be regarded as process-relevant. Therefore, interpretability in hydrological modeling should move ***beyond feature attribution*** and incorporate behavioral validation, such as sensitivity analysis and counterfactual experiments[30], to establish consistency between explanation results, model behavior, and underlying hydrological processes[31].

In addition, existing interpretability approaches are largely confined to individual models and lack systematic evaluation[32] of whether the learned knowledge is transferable. If the key feature relationships or internal structures identified in one model cannot be reproduced or improve performance in other model frameworks, such explanations are more likely to reflect model-specific parameterization rather than generalizable hydrological knowledge. From a scientific perspective, meaningful explanations should hold across different model architectures and be validated beyond a single model[33]. Therefore, cross-model validation is a crucial step in determining whether the extracted knowledge possesses true scientific value.

In summary, current interpretability studies in deep learning-based streamflow prediction face three major challenges: (1) **lack of verifiability**, making it difficult to quantitatively validate explanation results; (2) **lack of behavior-level consistency evaluation**, particularly the absence of sensitivity analysis to assess whether model responses align with hydrological principles; and (3) **lack of transferability assessment**, leaving it unclear whether the learned knowledge is model-independent and generalizable. These limitations collectively constrain the scientific interpretability and practical reliability of existing approaches.Notably, other fields have increasingly shifted toward rigorous and verification-oriented interpretability paradigms, such systematic efforts remain limited in hydrological modeling.

To address the three major challenges, this study proposes a novel framework, Sparse Transformer-based Explainable Hydrology (STEH), which enables full-chain interpretability from inputs through internal model structure to outputs. Specifically, (1) A sparse circuit discovery mechanism is introduced via learnable gating across input features, attention heads, and neurons, allowing the model to identify compact and functionally meaningful computational pathways (e.g., evaporation, precipitation–runoff transformation); (2) circuit-level ablation experiments are conducted to quantitatively verify the contribution of individual components; (3) internal dependency structures are analyzed to reveal information flow and align model representations with hydrological processes; (4) sensitivity and counterfactual analyses are performed to evaluate whether model responses are physically consistent; and (5) cross-model transfer experiments are designed by incorporating the extracted structural knowledge into a Random Forest benchmark, thereby assessing the transferability of learned knowledge.

The main contribution of this study lies not only in the development of an interpretable deep learning framework, but also in establishing a systematic validation pathway for interpretability in hydrological modeling. By integrating structural analysis, behavioral verification, and cross-model evaluation, this work advances the understanding of interpretability from merely explaining model outputs to rigorously assessing whether such explanations are reliable, physically meaningful, and transferable. In doing so, it provides preliminary insights into a fundamental question: ***what kind of hydrological knowledge is learned, and whether it is verifiable and transferable?***

2 Methods

2.1 Framework Overview

The overall workflow of the proposed Sparse Transformer-based Explainable Hydrology (STEH) framework is illustrated in Fig. 1. The framework provides an end-to-end interpretability pipeline for deep learning-based streamflow prediction, enabling systematic analysis from hydrological input variables, through internal neural structures, to prediction outputs.

The STEH framework consists of three main components: (1) sparse circuit discovery, (2) circuit verification and refinement, and (3) physical interpretation. Together, these components aim to reveal how deep learning models process hydrological information and whether the learned representations correspond to meaningful hydrological processes.

First, sparse circuit discovery is performed by introducing learnable gating mechanisms into the Transformer architecture. Gates are applied to input variables, attention heads, and feed-forward neurons, allowing the model to automatically identify a compact set of critical computational units during training. Through sparsity-inducing optimization, redundant structures are gradually suppressed, resulting in a sparse computational circuit responsible for streamflow prediction.

Second, circuit verification and refinement are conducted through systematic ablation experiments. Individual computational units—including input nodes, attention heads, and neurons—are selectively removed to measure their structural contribution to prediction performance. Units with negligible influence are discarded, yielding a compact core predictive circuit that preserves the essential functional structure of the model.

Third, the extracted core circuit is further analyzed to understand internal information flow and structural dependencies. Interaction analysis is used to identify cooperative or redundant relationships between model components. In addition, linear probing techniques are applied to hidden representations to evaluate whether internal states encode physically meaningful hydrological variables.

Finally, the interpretability results are validated from a hydrological perspective. Based on the structural knowledge extracted from the neural network, modifications are introduced to a Random Forest benchmark model to examine whether the discovered hydrological knowledge can improve data-driven hydrological simulations.

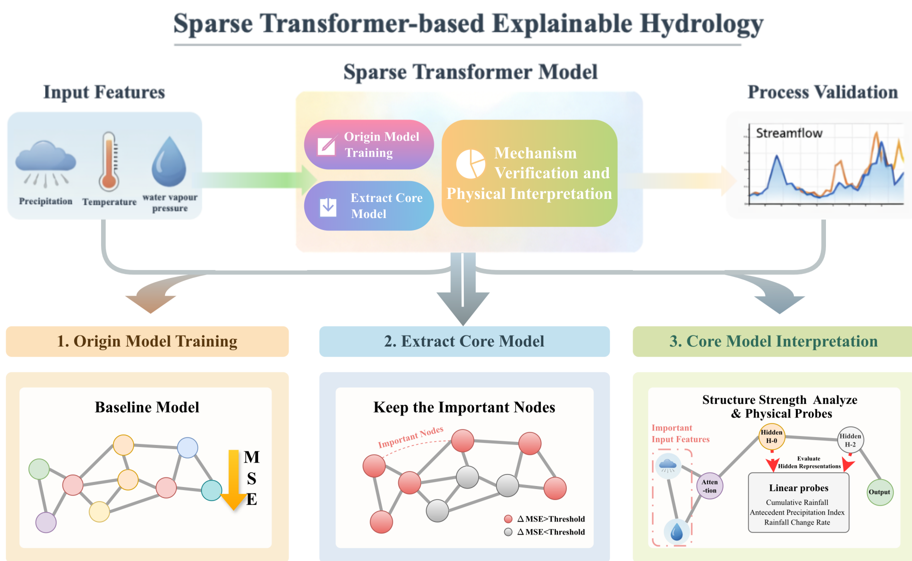

Fig. 1 Overall architecture of the proposed STEH framework

2.2 Sparse Transformer Architecture

Transformer Formulation

The Transformer architecture has recently been introduced into hydrological time-series modelling because its self-attention mechanism can effectively capture long-range temporal dependencies in hydro-meteorological sequences[34]. In streamflow prediction, antecedent rainfall, temperature, and soil moisture conditions jointly influence discharge responses across multiple time lags, making long-term dependency modelling particularly important[35].

Let $\mathbf{X} \in \mathbb{R}^{L \times F}$ denote the input sequence with length L and feature dimension F. The self-attention operation is defined as[36]

$$\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\top}}{\sqrt{d_k}}\right)\mathbf{V}, \tag{1}$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ represent the query, key, and value matrices derived from the input representations, and d_k denotes the key dimension used for scaling.

Hidden representations are updated through stacked Transformer encoder blocks[37]:

$$\mathbf{H}^{(l+1)} = \mathrm{Block}^{(l)}(\mathbf{H}^{(l)}), \quad l = 0, \ldots, N-1, \tag{2}$$

where $\mathbf{H}^{(l)}$ represents the hidden state of layer l, and N denotes the number of encoder layers.

Structured Gating Mechanism

To enable circuit-level interpretability, structured gating mechanisms are introduced at three levels of the network architecture: input variables, attention heads, and feed-forward neurons[38]. The effect of gate-based circuit selection is illustrated in Fig. 2(e–f).

For the input layer, the gated representation is defined as

$$\tilde{\mathbf{X}} = \mathbf{X} \odot \mathbf{m}_{\text{in}}, \tag{3}$$

where $\mathbf{m}_{\text{in}} \in \{0,1\}^F$ is a binary mask indicating whether each input feature is retained.

Similarly, head-level and neuron-level masks are defined for each Transformer layer, denoted by $\mathbf{m}_{\text{head}}^{(l)}$ and $\mathbf{m}_{\text{neu}}^{(l)}$, respectively. These masks control the activation of attention heads and feed-forward neurons[39], enabling structured sparsification of the model.

Sparse Training Strategy

Training is conducted in three consecutive stages to stabilize optimization and avoid premature pruning of potentially important connections[40].

Stage I (Origin Model Training). The model is first trained without sparsity constraints to establish a stable predictive baseline (Fig. 2(a–b)).

Stage II (Weight Sparsification). Top-K sparsity is progressively imposed on weight tensors with an annealed sparsity schedule[41] (Fig. 2(c–d)):

$$\rho_t = \rho_{\max} - (\rho_{\max} - \rho_{\min}) \left(1 - \frac{t}{T}\right)^{\gamma}, \tag{4}$$

where $\rho_{\min}$ and $\rho_{\max}$ denote the initial and final sparsity ratios. t represents the pruning iteration, T is the total number of steps over which sparsification is applied, and γ is a shape parameter that controls the annealing schedule. Specifically, larger γ values result in a slower increase in sparsity at early stages and a more rapid transition near the end, while smaller γ leads to a more uniform sparsification process.

Stage III (Mask Learning[42]). Learnable gates are optimized with sparsity regularization (Fig. 2(e–f)):

$$\begin{cases} \mathcal{L} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{gate}}, \\ \mathcal{L}_{\text{task}} = \dfrac{1}{M} \displaystyle\sum_{n=1}^{M} (\hat{y}_n - y_n)^2, \\ \mathcal{L}_{\text{gate}} = \lambda_{\text{gate}} \dfrac{1}{|\mathcal{G}|} \displaystyle\sum_{g \in \mathcal{G}} p_g. \end{cases} \tag{5}$$

where M is the number of training samples in a mini-batch, $\mathcal{L}_{\text{task}}$ is the mean squared error loss, $\mathcal{L}_{\text{gate}}$ is the overall gate sparsity regularizer, $\mathcal{G}$ denotes all learnable gates. We consider two types of gates: input-level gates and node-level gates, corresponding to *input* and *head_neu*, respectively, and p_g is the activation probability of gate g.

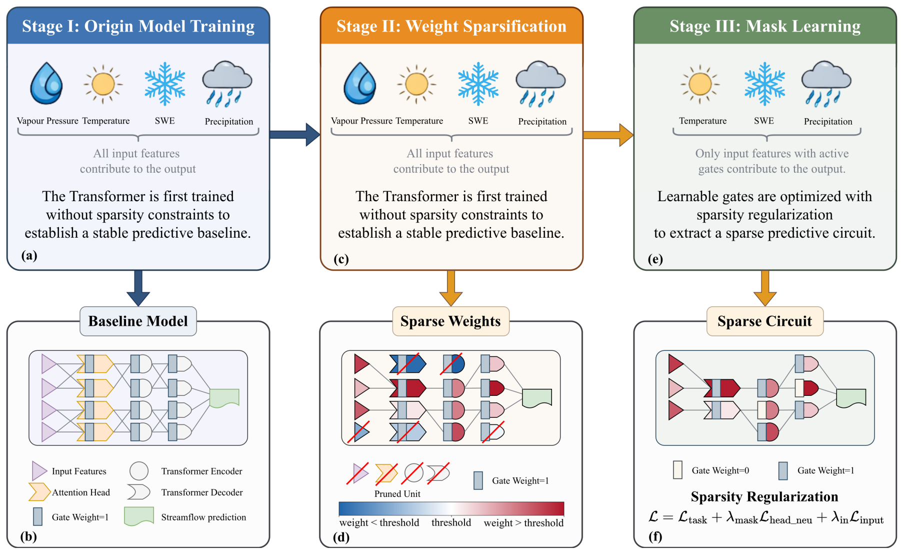

Fig. 2 Three-stage sparsification framework for predictive circuit discovery in Transformer. The model is trained through three stages to progressively transform a dense Transformer into a sparse predictive circuit. (a–b) Stage I: Origin model training, where the Transformer is trained without sparsity constraints and all input features contribute to the prediction. (c–d) Stage II: Weight sparsification progressively prunes redundant connections according to the sparsity schedule. (e–f) Stage III: Mask learning optimizes learnable gates so that only a subset of input features contributes to the final prediction. For clarity, only a subset of connections between layers is illustrated in the figure, while the actual model remains fully connected, and decoder attention layers are omitted for simplicity. Note that the proposed method does not explicitly modify the network architecture; sparsity is achieved by setting weights or gates to zero, while the visual removal of units in the figure is only for illustrative purposes. The number of input variables shown in the figure is fewer than the actual input feature set; this simplification is adopted solely for visual clarity.

After training, a compact core predictive circuit is extracted by thresholding gate probabilities.[43] Let $p_i = \sigma(z_i)$ denote the activation probability of gate i. A unit is retained if

$$a_i = \mathrm{I}(p_i > \tau), \tag{6}$$

where τ is the pruning threshold. Applying this rule to all gates yields a compact model containing only the most important computational pathways.

2.3 Interpretability Analysis

Ablation-based Importance

To quantify the causal importance of individual computational units, targeted ablation is applied to each candidate unit u [44]. The importance score is defined as

$$\Delta\mathrm{MSE}_u = \mathrm{MSE}(\mathrm{ablate}(u)) - \mathrm{MSE}(\mathrm{base}). \tag{7}$$

Unlike gradient-based attribution methods, this intervention-based measurement directly evaluates the causal contribution of each component by observing the performance change after structural removal[45].

Interaction Analysis

To analyze dependencies between components, pairwise interaction scores are defined as

$$I_{ij} = \Delta_{ij} - \Delta_i - \Delta_j. \tag{8}$$

Positive values indicate synergistic interactions, whereas negative values suggest redundancy or functional substitution.

Top-K Faithfulness

To evaluate the trade-off between circuit compactness and predictive fidelity, a top-K faithfulness analysis is conducted by retaining only the most important units and measuring the resulting performance degradation[46].

Physical Probing

To examine whether hidden representations encode physically meaningful hydrological information, linear probes are applied to hidden states[47].

Let $\mathbf{h}_n \in \mathbb{R}^D$ denote the hidden representation of sample n. The probe model is defined as

$$\hat{y}_n^{\mathrm{phys}} = \mathbf{w}^\top \mathbf{h}_n + b, \tag{9}$$

where $\mathbf{w}$ and b are parameters estimated using least squares regression.

Probe performance is evaluated using the coefficient of determination (R^2). To avoid probe overfitting, the probe model is intentionally restricted to a simple linear form without hidden layers.

Representative hydrological variables such as cumulative rainfall[48], antecedent precipitation index (API)[49], and rainfall change rate are used as probe targets. Let P_t denote precipitation at day t, and let W be the look-back window length. Their definitions are

$$R_t^{\mathrm{cum}}(W) = \sum_{k=0}^{W-1} P_{t-k}, \tag{10}$$

$$\mathrm{API}_t(W, \kappa) = \sum_{k=0}^{W-1} \kappa^k P_{t-k}, \qquad 0 < \kappa < 1, \tag{11}$$

$$R_t^{\mathrm{chg}} = \frac{P_t - P_{t-1}}{P_{t-1} + \varepsilon}, \tag{12}$$

where κ is a decay factor controlling the memory of antecedent rainfall and ε is a small constant to avoid division by zero.

2.4 Evaluation Metrics

Model evaluation is conducted from three perspectives: predictive accuracy, explainability properties, and hydro-climatic diagnostics.

Predictive accuracy is measured using standard hydrological metrics including RMSE[50], MAE[51], and NSE[52]. Explainability-related metrics include ablation importance (ΔMSE, Eq. 7), interaction score (I_{ij}, Eq. 8), and top-K faithfulness[46] trends. Structural sparsity statistics are also reported to quantify circuit compactness.

Hydro-climatic diagnostics are used to characterize the climatic regimes of catchments. Let $\bar{P}$, $\overline{\mathrm{PET}}$, and $\bar{Q}$ denote the long-term mean annual precipitation, potential evapotranspiration, and streamflow, respectively, and let P_m ($m = 1, \ldots, 12$) be the mean monthly precipitation with $\bar{P}_{\mathrm{ann}} = \sum_{m=1}^{12} P_m$.

The **aridity index** (AI)[53] describes the balance between water supply and atmospheric demand:

$$\mathrm{AI} = \frac{\bar{P}}{\overline{\mathrm{PET}}}. \tag{13}$$

Values $\mathrm{AI} < 1$ indicate arid conditions, while $\mathrm{AI} > 1$ indicates humid conditions.

The **Streamflow coefficient** (C_r)[54] represents the fraction of precipitation that becomes streamflow:

$$C_r = \frac{\bar{Q}}{\bar{P}}. \tag{14}$$

The **precipitation seasonality index** (SI)[55] quantifies the degree of seasonal concentration of rainfall:

$$\mathrm{SI} = \frac{1}{\bar{P}_{\mathrm{ann}}} \sum_{m=1}^{12} \left| P_m - \frac{\bar{P}_{\mathrm{ann}}}{12} \right|, \tag{15}$$

where $\mathrm{SI} = 0$ indicates perfectly uniform monthly distribution and larger values correspond to stronger seasonal concentration. Together, these three diagnostics are used to stratify stations by hydro-climatic regime and to contextualize model performance across different environmental conditions.

3 Data

This study uses the CAMELS-US dataset (https://doi.org/10.1016/0022-1694(89)90184-4), which provides daily hydro-meteorological time series for minimally disturbed catchments across the contiguous United States [49]. The dataset is selected for three reasons: (1) the catchments are weakly affected by direct human regulation, which is suitable for identifying robust hydrological knowledge; (2) the long and standardized daily forcing-streamflow records are well suited to deep-learning time-series training and cross-basin comparison [56]; and (3) the data organization is consistent with large-sample hydrology workflows, supporting reproducible model evaluation [29]. To ensure temporal consistency across stations, the common period from 1980-01-01 to 2014-12-31 is selected for analysis.

Because the availability of streamflow records varies across stations, a unified quality-control procedure is applied within this period. Stations with more than 100 missing daily values in the target streamflow series are excluded, whereas stations with at most 100 missing days are retained and the missing values are filled using linear interpolation. After preprocessing, 521 stations remain for subsequent analysis. All retained stations provide multi-year continuous records over the common window (well beyond 20 years), ensuring sufficient sample size for deep-learning training.

The model input variables are summarized in Table 1. For each station, seven daily meteorological forcing variables from Daymet are used, and daily streamflow is additionally incorporated to represent antecedent hydrological state knowledge.

Table 1 Selected Input Meteorological and Hydrological Variables

Group	Feature name	Unit
Energy	day length (dayl)	s day^{-1}
	shortwave radiation (srad)	W m^{-2}
Temperature	maximum air temperature (tmax)	°C
	minimum air temperature (tmin)	°C
Atmospheric moisture	water vapour pressure (vp)	Pa
Precipitation	precipi tation (prcp)	mm day^{-1}
Snow	snow water equivalent (swe)	mm
Runoff	streamflow (Q)	mm day^{-1}

The prediction task is formulated as sequence-to-one regression. A sliding window of $L = 15$ days is used to predict streamflow at the next time step.

All features are standardized using training-set statistics. Data are split chronologically into training (70%), validation (15%), and test (15%) subsets.

The model is implemented in Python 3.10 using PyTorch, and all experiments are conducted on a Linux server equipped with NVIDIA CUDA-enabled GPUs. The three-stage training pipeline (dense pre-training, weight sparsification, mask learning) shares a single AdamW optimiser with weight decay. Hyperparameters are tuned via Bayesian optimisation[57], with the search space covering learning rate (10^{-4}–10^{-2}), batch size (32–256), hidden size (16–256), number of layer (1–16), target weight sparsity (0.7–

0.95), target activation sparsity (0.6–0.9), and mask regularisation strengths λ_{mask} and λ_{in} (10^{-5}–10^{-3}). The objective of Bayesian optimisation is the validation NSE, and the optimal configuration is then applied consistently across the three training stages and all subsequent explainability analyses reported in this study. Additional details can be found in Supplementary Notes 2.1.

4 Results

4.1 Robust Predictive Performance Across Hydrological Stations

Before interpreting the internal knowledge structures of the model, it is essential to verify that the model achieves reliable predictive performance across hydrological stations[58]. In this study, predictive performance is evaluated using the NSE, with a threshold of NSE > 0.5 indicating satisfactory model performance[59].

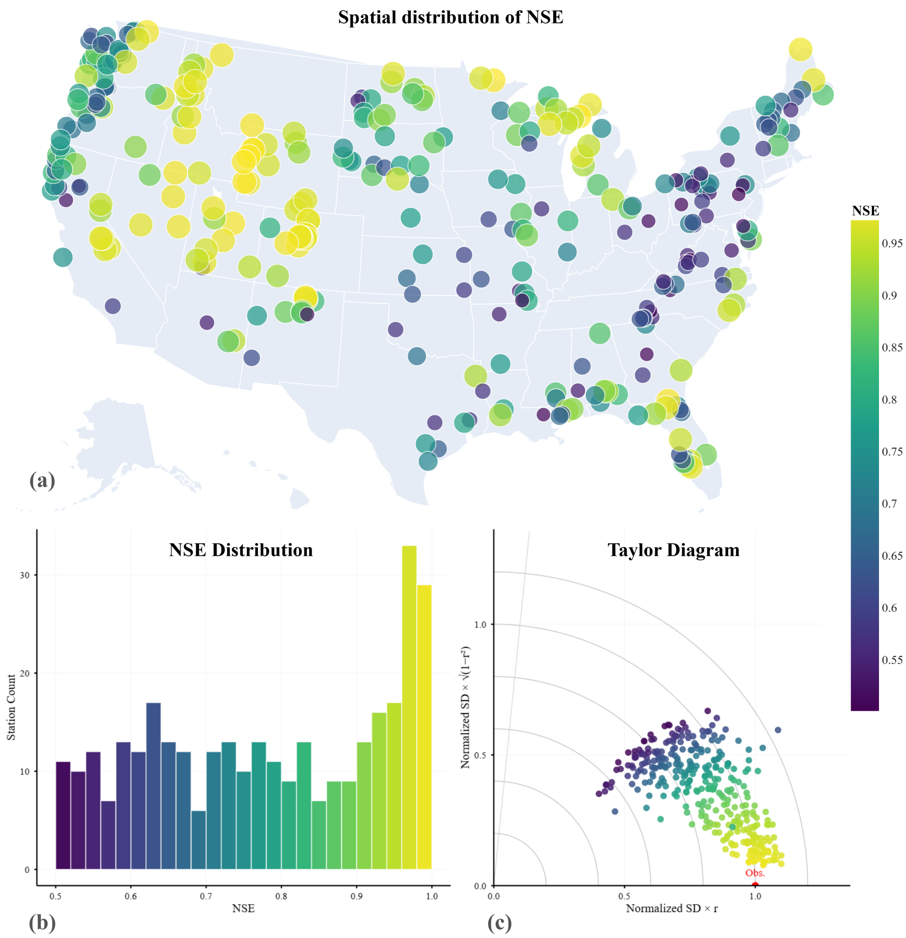

Fig. 3 **Model performance across 521 streamflow gauging stations.** (a) Spatial distribution of NSE, with warmer colors indicating higher predictive skill. (b) Distribution of NSE values across all stations, showing that the majority achieve satisfactory performance (NSE $>$ 0.5). (c) Taylor diagram summarizing model performance in terms of correlation, normalized standard deviation, and centered root-mean-square error relative to observations. Overall, the results demonstrate that the model provides robust predictions for a large proportion of stations, while performance varies across regions, reflecting differences in hydrological conditions.

As shown in Fig. 3, the STEH framework demonstrates robust performance across a large proportion of stations, despite being specifically designed to enhance interpretability rather than maximize predictive accuracy. Specifically, 327 out of 521 stations (approximately 62.8%) achieve NSE values greater than 0.5, indicating satisfactory predictive skill in the majority of cases[50].

This result suggests that the model is capable of capturing the dominant hydrological dynamics in most basins[52], thereby providing a reliable foundation for subsequent interpretability analysis[58]. Consequently, the insights derived from the STEH framework are grounded in meaningful predictive behavior rather than artifacts of poor model performance[56].Based on this reliable predictive foundation, we next investigate the internal circuit structures of the model.

4.2 Interpretable Circuit Structures Across Hydrological Stations

Given the large number of stations (521), it is impractical to visualize the learned circuits for all stations individually. Therefore, a representative sampling strategy is adopted to systematically examine circuit structures under different predictive conditions.

Specifically, stations are categorized into three performance groups[59] based on the NSE. One representative station is selected from each category, as summarized in Table 2.

Table 2 Representative stations across NSE performance categories

Performance Category	NSE Range	Station ID	NSE
Very Good	NSE > 0.75	09066300	0.9905
Good	$0.65 <$ NSE ≤ 0.75	01439500	0.7366
Satisfactory	$0.50 <$ NSE ≤ 0.65	03504000	0.6217

First, a representative inland station with very good performance (Station ID: 09066300, *Middle Creek near Minturn, CO.*, 39.65°N, 106.38°W, NSE = 0.9905) is selected to analyze the extracted core circuit (Fig. 4). The station is located in a high-altitude inland basin in Colorado, USA (approximately 3251 m), where hydrological processes are primarily controlled by temperature-driven dynamics and internal catchment routing[60].

From the circuit structure (Fig. 4a), it can be observed that the model retains a clear and stable information flow pathway after strong sparsification. Among the input variables, streamflow (Q) and maximum temperature (Tmax) dominate, with significantly stronger connections than DayL. Rather than indicating only variable importance, this structure reveals a hydrological pattern in which antecedent catchment state (Q as a storage proxy) and temperature jointly regulate runoff release. In this station, STEH therefore identifies a state-modulated, temperature-sensitive flow-generation pathway that is consistent with high-altitude processes such as snowmelt timing and evapotranspiration control[43]. Within the hidden layers, information propagates through a small number of key nodes and converges to the output, forming a compact and low-redundancy computational pathway.

Second, from the training dynamics (Fig. 4a, lower right), as the number of retained nodes (K) increases, NSE rapidly improves to near 1 and stabilizes when K remains relatively small (K ¡ 100), while MSE decreases sharply and stays low. Notably, the model achieves performance comparable to the full model with only a small subset of nodes. This demonstrates that the extracted sparse circuit preserves the essential predictive capability of the original model, confirming its functional fidelity[40]. Meanwhile, it also indicates that the sparsity-inducing training successfully identifies and retains the most critical computational units, effectively removing redundant structures and yielding a compact yet accurate representation of the original network[45].

Furthermore, the interaction analysis (Fig. 4b) reveals strong dependencies among input variables. In particular, Tmax exhibits the strongest interaction with Q, followed by the interaction between Q and DayL, while DayL shows relatively weak interactions overall. This supports a process-level interpretation: thermal forcing does not act independently, but modulates discharge response through antecedent-state conditioning, consistent with energy-state coupling in mountain basins[61].

Finally, the contribution analysis (Fig. 4c) shows a clear imbalance among input variables. Streamflow (Q) contributes the most, followed by Tmax, while DayL has the smallest contribution. Combined with the circuit and interaction evidence, this indicates that the dominant hydrological pattern at this site is a state-dependent runoff pathway with secondary temperature regulation, rather than a uniformly distributed multi-driver response.

Overall, for this very good inland station, the extracted core circuit is highly sparse while maintaining strong predictive capability[46]. More importantly, STEH recovers a coherent hydrological mode: antecedent state and thermal forcing jointly govern streamflow response, indicating that the learned sparse circuit encodes hydrologically relevant knowledge rather than only statistical attribution.

For a representative station with good performance (Station ID: 01439500, *Bush Kill at Shoemakers, PA*, 41.09°N, 75.04°W, NSE = 0.7366), the extracted circuit exhibits a moderately sparse structure (Fig. 5). Compared to the inland very good station, more input variables contribute to the prediction, including precipitation (P), streamflow (Q), and radiation-related variables. The circuit shows a more distributed information flow[45], indicating that multiple sub-processes are active simultaneously. Interaction analysis further suggests that Q remains dominant but is strongly coupled with precipitation, which can be interpreted as an event-response pathway where rainfall inputs are filtered by antecedent wetness and then transmitted to discharge[62]. Overall, the extracted circuit still preserves the main predictive capability, though with increased complexity and reduced sparsity.

For a representative station with satisfactory performance (Station ID: 03504000, *Nantahala River near Rainbow Springs, NC*, 35.13°N, 83.62°W, NSE = 0.6217), the extracted circuit becomes less structured and more diffuse (Fig. 6). Although streamflow (Q) still dominates the prediction, other variables such as radiation (SRAD) and DayL show weaker and less consistent contributions. The interaction patterns are less pronounced, suggesting that no single robust hydrological pathway is cleanly separated from competing signals. In addition, the circuit retains more redundant connections, indicating that the sparsity-inducing process is less effective in isolating a compact predictive core[43]. This reflects the reduced model performance and highlights the in-

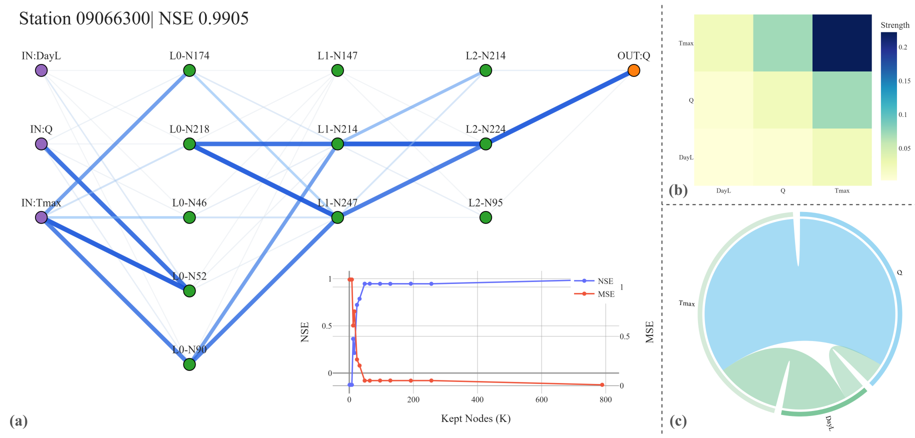

Fig. 4 **Extracted sparse circuit and interpretability analysis for a representative inland station with very good performance (NSE = 0.9905).** (a) Core circuit structure and training dynamics, where lighter-colored edges indicate weaker contributions and less important pathways. (b) Interaction strength between input variables, illustrating pairwise dependencies. (c) Contribution of each input variable to streamflow prediction.

creasing difficulty of extracting clear and physically consistent hydrological knowledge in lower-performing cases[63].

Comparing the three representative stations, a clear pattern emerges: as predictive performance decreases, the extracted circuits become less sparse and more diffuse. High-performing stations exhibit compact and structured pathways that map to identifiable hydrological modes, while lower-performing stations show increased redundancy, weaker interaction coherence, and less stable knowledge-level interpretation.

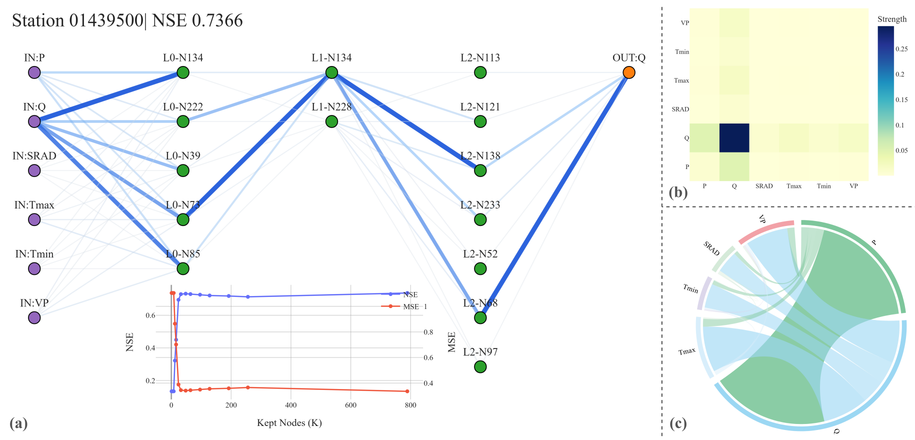

Fig. 5 Extracted sparse circuit for a representative station with good performance (NSE = 0.7366).

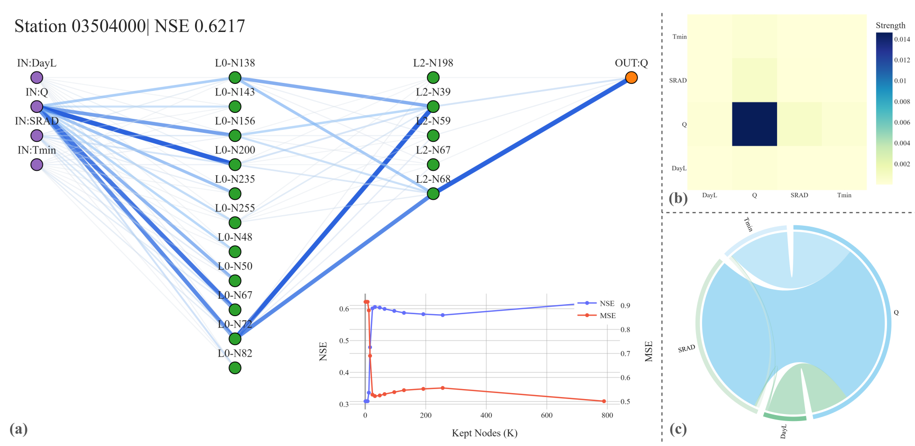

Fig. 6 Extracted sparse circuit for a representative station with satisfactory performance (NSE = 0.6217).

5 Discussion

5.1 Parsimony and Physical Interpretability Are Prevalent Across Models

Interestingly, our findings reveal a trend that contrasts with observations reported in some prior studies[64, 65], where increased model complexity has been associated with improved predictive performance. As shown in Table 3, models with higher predictive skill (NSE $\geq$ 0.5) exhibit significantly lower structural complexity compared to lower-performing counterparts, as indicated by a reduced number of nodes (45.77 vs. 56.35, $p = 5.00 \times 10^{-5}$). This suggests that increased model complexity does not necessarily translate into improved performance in this context; instead, more parsimonious architectures may facilitate better generalization[63]. One possible explanation is that overly complex models introduce redundant representations or noise, which can degrade predictive robustness in hydrological applications[66].

Several factors may explain why additional complexity can reduce performance in this setting. First, a larger circuit can absorb site-specific noise and transient correlations, which weakens out-of-sample robustness under hydro-climatic variability. Second, when many correlated predictors are represented through redundant internal paths, the model may dilute dominant controls (e.g., precipitation-state coupling) across multiple weak pathways, reducing effective signal-to-noise ratio. Third, in sparse-training regimes, over-parameterized structures increase optimization burden and make it harder to isolate a stable core predictive subnetwork, so pruning removes less redundancy and retains more unstable interactions. Collectively, these effects can produce a complexity penalty rather than a complexity benefit.

Table 3 Structural complexity across NSE-based performance groups

Group	Stations	Mean Nodes	Median Nodes
NSE $<$ 0.5	194	56.35	56.50
NSE $\geq$ 0.5	327	45.77	42.00
All	521	49.71	49.00

Note: A statistically significant difference is observed between groups ($p = 5.00 \times 10^{-5}$).

From the perspective of physical interpretability, the probe analysis (Table 4) indicates that key hydrological variables—including CUM, API, and CHG consistently encoded in the learned representations across models[47]. Specifically, no statistically significant differences are observed in probe performance between NSE-based groups (all $p > 0.05$). This implies that the ability to capture physically meaningful signals is not restricted to high-performing models[22], but rather is a common characteristic across the model population.

Taken together, these results suggest that parsimony and physical interpretability are not competing objectives. Instead, models with simpler structures are capable of achieving strong predictive performance while still encoding essential hydrological

Table 4 Probe performance under different NSE conditions

Probe	Group	Mean	Median	p-value
R^2_{cum}	NSE < 0.5	0.8746	0.9639	0.4341
	NSE ≥ 0.5	0.8915	0.9617	
	All	0.8851	0.9621	
R^2_{api}	NSE < 0.5	0.8795	0.9682	0.5676
	NSE ≥ 0.5	0.8917	0.9651	
	All	0.8871	0.9663	
R^2_{chg}	NSE < 0.5	0.3791	0.4267	0.1563
	NSE ≥ 0.5	0.4054	0.4406	
	All	0.3955	0.4342	

Outliers were excluded using the 1.5 IQR criterion within each group. Results remain consistent without filtering.

knowledge related to streamflow generation. This result motivates the next question: *if such knowledge is broadly present across models, what is its concrete content and spatial organization across basins?*

5.2 Learned Core Circuits Reflect Hydrological Knowledge Across Basins

To answer that question, we examine the spatial organization of the learned core interaction circuits. The results show that the model captures structured and regime-dependent hydrological knowledge across diverse basins[56]. As shown in Fig. 7(a), the dominant interaction in most basins is between precipitation and discharge (P–Q), reflecting the fundamental role of precipitation as the primary driver of streamflow generation[67]. Importantly, this is not only a feature-attribution result; it indicates a recurrent hydrological mode in which rainfall forcing dominates the event-scale runoff pathway across many catchments.

In contrast, a subset of basins is characterized by strong interactions between lagged discharge and target discharge (Q–Q_{target}), suggesting pronounced memory effects[4]. In these basins, streamflow dynamics are more strongly influenced by antecedent conditions than by immediate precipitation inputs, which is typically associated with limited precipitation or strong storage capacity. This pattern can be interpreted as a storage-controlled response mode, where antecedent wetness and catchment retention regulate subsequent flow release.

Temperature-related interactions (T_{max}–Q and T_{min}–Q) are primarily observed in inland and high-elevation regions. These patterns indicate that temperature plays a critical role in regulating streamflow generation in such environments, not only through snowmelt processes but also via evaporation and other energy-limited controls[3]. In addition, interactions involving shortwave radiation (SRD) emerge in some basins, further highlighting the importance of energy inputs[2] in modulating hydrological

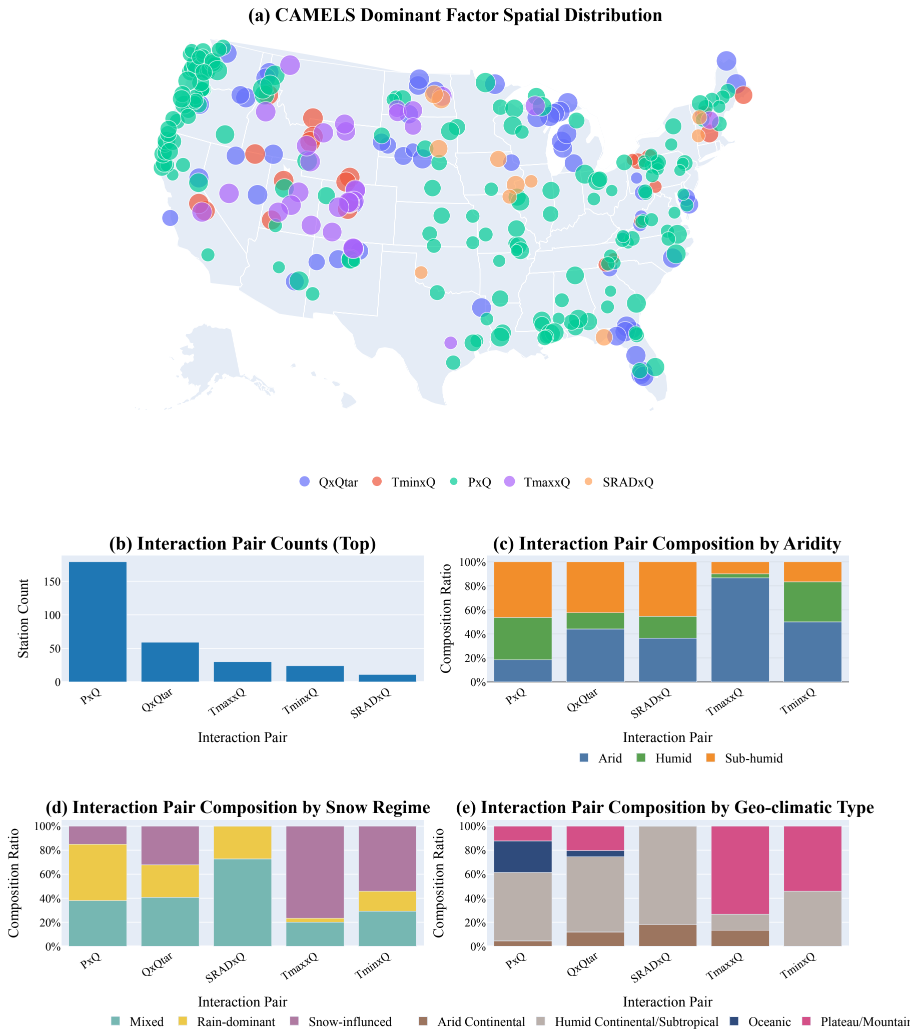

Fig. 7 (a) Spatial distribution of dominant interaction factors across basins. (b) Frequency of top interaction pairs. (c) Interaction composition under different aridity conditions. (d) Interaction composition across snow regimes. (e) Interaction composition across geo-climatic types.

responses. Together, these signals correspond to an energy-constrained runoff mode, distinct from precipitation-dominated systems.

These patterns remain consistent across different hydro-climatic conditions, as illustrated in Fig. 7(b–e). Precipitation-driven interactions dominate across most basins, whereas temperature-related and lagged discharge interactions become more prominent under specific environmental settings. In arid and semi-humid regions, temperature and memory-related interactions are more frequently observed[3], indicating stronger energy constraints and persistence effects. In contrast, humid basins are largely governed by precipitation-driven dynamics[67].

Similarly, temperature-related interactions[2] are concentrated in snow-influenced basins, consistent with snowmelt-controlled streamflow processes, while precipitation-driven interactions dominate in rain-fed systems[68]. Across different geo-climatic types, distinct interaction circuits exhibit clear regional preferences, suggesting that the learned structures adapt to local environmental conditions.

Overall, these findings demonstrate that the learned core circuits are not arbitrary but systematically reflect regime-dependent hydrological patterns[69]. Thus, STEH moves the interpretation target from ”which variable is important” to ”which process pathway is operating,” providing structural evidence that the model internalizes physically meaningful knowledge beyond simple variable associations. However, structural coherence alone is insufficient: the extracted knowledge should also produce physically consistent behavior under controlled perturbations.

5.3 Accurate Volume Response but Limited Sensitivity to Temporal and State Perturbations

Following the structural analysis above, we next test whether the extracted hydrological knowledge corresponds to realistic model behavior. As shown in Fig. 8, the counterfactual analysis reveals that the model exhibits physically meaningful responses under different types of perturbations[31]. For magnitude-based interventions, particularly rainfall scaling (Fig. 8b), the model produces directionally consistent and approximately linear changes in streamflow volume across stations. As precipitation increases (or decreases), the predicted streamflow volume correspondingly increases (or decreases), while prediction error remains largely unchanged. This indicates that the model has successfully learned a stable and physically consistent mapping between precipitation and streamflow generation, allowing it to respond reliably to changes in input water magnitude without compromising predictive accuracy[29].

Under rainfall pattern redistribution (Fig. 8c), the total precipitation is conserved, and the predicted streamflow volume remains largely stable, consistent with mass conservation expectations[52]. However, structural differences emerge across redistribution modes. In particular, under the back-loaded configuration, a noticeable increase in streamflow volume is observed. This suggests that the model is sensitive to the temporal concentration of rainfall and produces stronger streamflow responses[56] when precipitation is concentrated toward later periods, reflecting an initial capability to capture rainfall-streamflow lag relationships. Meanwhile, the variation in prediction error across scenarios indicates that the model's utilization of temporal structure remains somewhat scenario-dependent.

For streamflow-state perturbations (Fig. 8d), moderate scaling of recent streamflow history leads to corresponding adjustments in predicted volume, while prediction error remains stable. This demonstrates that the model can effectively utilize state information while maintaining consistent predictive performance. As the scaling magnitude increases, however, the model response gradually saturates, with volume changes no longer increasing proportionally to the input. This attenuation behavior indicates that the model does not simply amplify streamflow magnitude linearly, but instead imposes constraints on non-structured or inconsistent input variations, thereby avoiding unrealistic over-amplification. Such behavior suggests that the model captures streamflow dynamics through interactions among multiple hydrological factors[70] rather than relying solely on absolute magnitude.

For temperature perturbations (Fig. 8e), the predicted streamflow volume shows a slight decreasing trend, with consistent direction across snow-dominated and rainfall-dominated catchments and no significant difference between groups. This aligns with hydrological expectations that warming generally reduces streamflow through enhanced evapotranspiration or phase-related processes. In contrast, prediction error exhibits significantly different sensitivities between the two regimes: snow-dominated catchments show substantially stronger responses to temperature changes than rainfall-dominated ones[3], and this difference is consistent across warming magnitudes (p ¡ 0.001). This suggests that the model not only captures the overall volumetric effects of temperature, but also differentiates between distinct hydrological knowledge patterns, reflecting sensitivity to process heterogeneity.

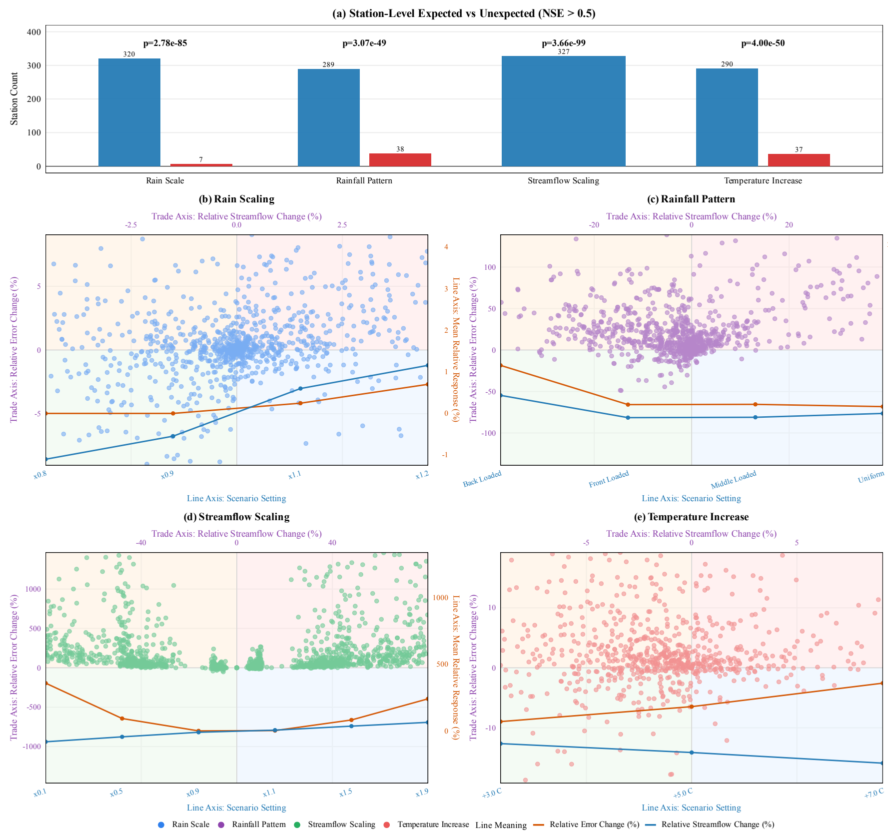

Fig. 8 **Counterfactual sensitivity analysis across intervention types.** (a) Station-level expected behavior statistics. (b) Rainfall scaling. (c) Rainfall pattern redistribution. (d) streamflow-state scaling. (e) Temperature perturbation. Blue lines denote relative changes in predicted streamflow volume, and orange lines denote relative changes in prediction error. Detailed experimental design and evaluation metrics are provided in Supplementary Information, Section 1.1

Overall, the model demonstrates robust responses to precipitation-driven volume changes and shows sensitivity to temporal structure and catchment state variations[71], indicating effective learning of key hydrological knowledge. At the same time, the differing response patterns across perturbation types suggest that the representation of more complex processes, such as temporal organization and state evolution, remains an area for further improvement. This behavioral consistency supports the validity of the extracted knowledge, and naturally leads to the final question: *can this knowledge be transferred to improve other model classes?*

5.4 Cross-Model Transfer Validates STEH-Derived Knowledge

To answer this final question, we evaluate transferability using an external model family. Specifically, we designed a comparative experiment in which STEH-derived structural information is incorporated into a conventional Random Forest model[72] and compared against a baseline model without such information. The key purpose is not only to improve another model, but to test whether the extracted structures represent model-independent hydrological knowledge that remains effective outside the Transformer architecture[73]. Furthermore, station-wise paired significance tests (paired t-test and Wilcoxon signed-rank test) demonstrate that the modified model significantly outperforms the baseline in a statistical sense, confirming the robustness of the observed improvements. Detailed descriptions of the model architecture, feature construction, and training and evaluation procedures are provided in Supplementary Note 1.2.

From an overall perspective, the modified model exhibits consistent performance improvements across all stations[35]. Specifically, the NSE increases from 0.760 to 0.767, accompanied by reductions in both MSE and MAE, indicating enhanced predictive accuracy and error control[51]. The station-wise comparison (Fig. 9(d)) shows that most points lie above the diagonal line, suggesting that the modified model outperforms the baseline in the majority of cases. In addition, the improvement is observed across a wide range of baseline performance levels (Fig. 9(e)), indicating that the benefit is not confined to specific subsets of stations. Together, these results indicate that the transferred structures retain predictive value in an independent model class, which supports the claim that STEH extracts non-model-specific hydrological knowledge rather than architecture-specific heuristics[74].

Across different hydro-climatic regimes, the most pronounced improvements are observed in arid regions. As shown in Fig. 9(a) and Table 5, inland arid and semi-arid areas exhibit predominantly positive performance gains. In the Arid group, the modified model consistently outperforms the baseline in terms of MSE, MAE, and NSE, indicating that the incorporation of structural information significantly enhances predictive capability under such conditions. This suggests that in environments where precipitation signals are weak and hydrological responses are governed by multiple interacting factors[75], the inclusion of additional structural features enables the model to better capture streamflow dynamics.

In humid regions, the modified model also achieves consistent improvements (NSE: 0.700 to 0.709), although the magnitude of improvement is relatively smaller. This can be attributed to the comparatively simpler hydrological processes in such regions[76], where streamflow generation is primarily controlled by a dominant driver. Under these

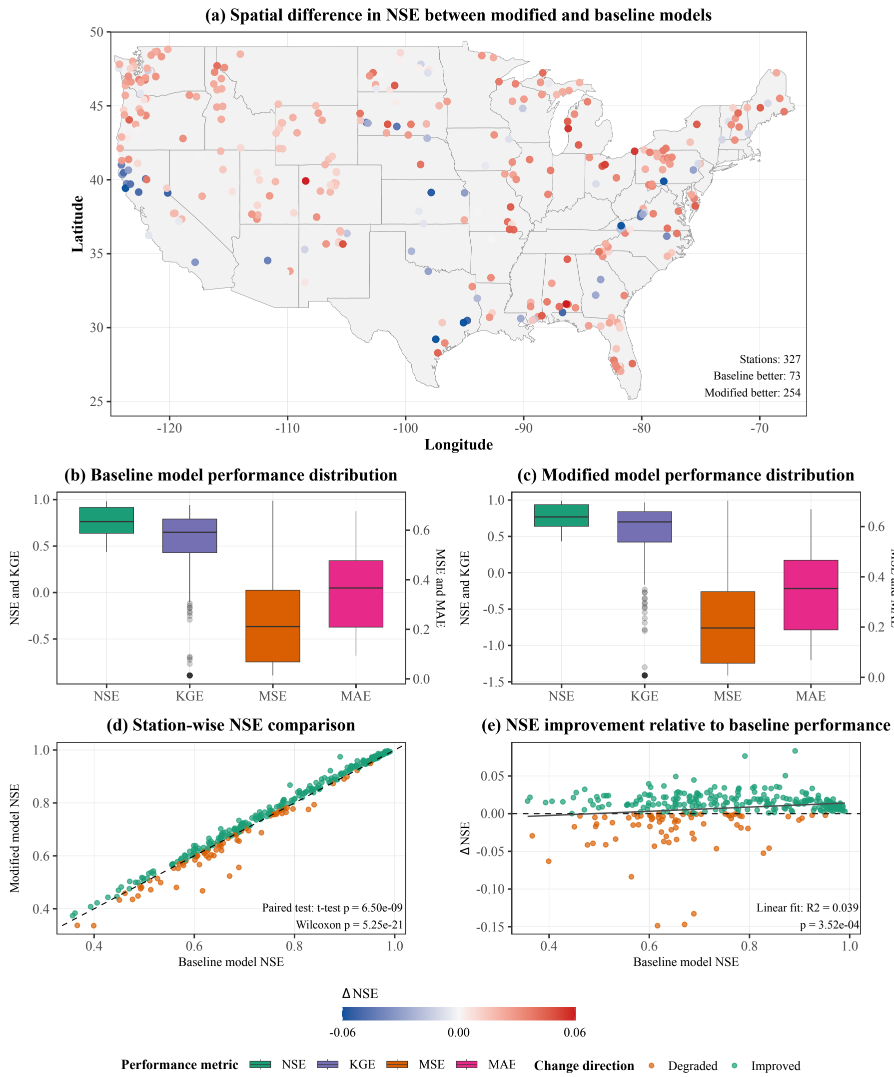

Fig. 9 **Performance comparison between baseline and STEH-informed Random Forest models.**(a) Spatial distribution of NSE differences (red: improved; blue: degraded).(b,c) Performance distributions of the baseline and modified models.(d) Station-wise NSE comparison (points above the diagonal indicate improvement).(e) NSE improvement relative to baseline performance.All results are based on stations with STEH model performance (NSE > 0.5).

Table 5 Performance comparison between baseline and STEH-informed modified Random Forest models across hydro-climatic groups

Model	Group	MSE	MAE	NSE
Baseline	Arid	0.140054	0.260355	0.841540
	Humid	0.324556	0.435797	0.700072
	Sub-humid	0.270571	0.385097	0.724154
	All Stations	0.237974	0.353715	0.759683
Modified	Arid	0.133525	0.248404	0.846403
	Humid	0.313798	0.425727	0.709464
	Sub-humid	0.259417	0.372654	0.733143
	All Stations	0.228561	0.342049	0.767310

All metrics are averaged across stations within each hydro-climatic group. Only stations with STEH model performance (NSE > 0.5) are included, as the extracted hydrological structures are derived from the STEH model and may become unreliable when its predictive performance is low.

conditions, the baseline model is already capable of capturing the main dynamics, leaving limited room for further improvement. Nevertheless, the continued reduction in error metrics indicates that structural information still contributes to improved fine-scale representation[77].

Overall, these findings demonstrate that the structural information extracted by STEH is not only interpretable but also scientifically verifiable through cross-model transfer. In particular, in regions characterized by complex or multi-factor-controlled processes, incorporating such structural knowledge can substantially improve model performance. More importantly, successful transfer to Random Forest provides an external validation step: the extracted knowledge is not tied to a single network design, but reflects portable hydrological regularities. This transfer result closes the full evidence chain of this section: hydrological knowledge is broadly encoded across models, its content can be structurally identified, its physical consistency can be behaviorally validated, and its non-model-specific validity can be verified through cross-model transfer.

6 Conclusion

Despite the rapid advancement of deep learning in hydrological prediction, model interpretability remains a fundamental challenge. Existing explanation approaches are often limited by insufficient verifiability, neglect of variable dependencies[22], and a lack of understanding of internal model structures[78]. As a result, it remains unclear whether high-performing models truly learn physically meaningful hydrological processes[58].

To address these challenges, this study proposes the Sparse Transformer-based[43] Explainable Hydrology (STEH) framework. The proposed framework establishes an end-to-end interpretability pipeline from inputs, through internal model structures, to outputs. By integrating structured sparsification, circuit-level ablation, variable interaction analysis[45], and representation probing, STEH provides a systematic approach to uncovering the internal decision-making knowledge structures of deep learning models in hydrological applications.

Overall, the main findings of this study can be summarized as follows.

First, this study provides a circuit-level perspective on deep learning models by explicitly revealing internal decision pathways and information flow. Compared with traditional feature attribution methods, this framework offers a structurally grounded and verifiable paradigm for interpretability.

Second, we identify a counterintuitive yet robust finding: simpler models tend to exhibit more compact internal circuits and stronger physical interpretability, whereas overly complex models often introduce redundant structures that fail to improve—and may even degrade—predictive performance. This result suggests that model parsimony not only benefits generalization but also facilitates the extraction of meaningful physical knowledge.

Third, we demonstrate that physically meaningful representations are consistently learned across models, regardless of predictive performance. Even lower-performing models retain substantial hydrological signals, indicating that physical knowledge is not exclusive to high-accuracy models but is inherently embedded within data-driven models.

Fourth, the STEH framework enables a reliable characterization of variable interactions from a structural perspective. The results reveal hydrologically consistent interaction patterns, including precipitation-dominated processes, temperature-driven dynamics in high-altitude regions, and memory effects associated with antecedent conditions. These interaction structures remain stable across diverse hydro-climatic regimes, suggesting that the learned patterns reflect real-world hydrological knowledge.

Fifth, sensitivity analysis shows that the model exhibits physically consistent responses to precipitation magnitude perturbations, while demonstrating limited sensitivity to temporal redistribution and state perturbations. This indicates that the model effectively captures dominant streamflow-generation knowledge but still lacks a complete representation of temporal organization and state evolution processes.

Sixth, the physical interpretability derived from STEH is further validated through its transferability to a Random Forest model. By incorporating STEH-derived structural information, the modified model achieves consistent and statistically significant performance improvements across stations. This demonstrates that the extracted knowledge is not only physically meaningful but also practically useful.

Nevertheless, several limitations should be acknowledged. First, although STEH improves structural interpretability, its results still cannot be interpreted as strict causal relationships[79]. This limitation arises from the intrinsic nature of deep learning models as statistical learning systems[80], where the extracted patterns primarily reflect statistical associations rather than true physical causality. Therefore, the findings derived from the model should be interpreted with caution when relating them to real-world physical processes.

Second, although this study reveals interaction structures among variables, their theoretical linkage to explicit physical process equations remains to be further explored. For example, the specific coupling relationships between variables, the pathways through which interactions operate, and the development of more controllable and interpretable circuit structures are not yet fully characterized[81]. Although the current framework enforces sparsity to simplify internal structures, the analysis is still largely limited to the relationship between input variables and the final hidden representations, leaving the information evolution within intermediate high-dimensional feature spaces insufficiently explored.

In addition, the validation of the extracted "physical knowledge" in this study is still based on machine learning models rather than explicit process-based hydrological models[82]. This may limit the physical rigor of the conclusions, and future work should incorporate process-driven models to further verify the findings.

In summary, this study demonstrates that deep learning models for hydrological prediction do encode physically meaningful information that can be systematically extracted through structural analysis. The proposed STEH framework not only provides a new paradigm for bridging black-box prediction and physical understanding, but also offers a promising pathway for feeding back into the improvement of traditional and physics-based models. These findings suggest that the role of deep learning in hydrology may extend beyond accurate prediction toward serving as a powerful tool for knowledge discovery and advancing the integration of data-driven and process-based modeling.

CRediT authorship contribution statement

Jiachen Kong: Conceptualization, Methodology, Software, Writing – original draft. **Zhaocai Wang**: Methodology, Software, Validation, Supervision, Writing – review & editing. **Zhaoyang Zhu**: Data curation, Writing – original draft.

Data availability

The data that support the findings of this study are publicly available. The streamflow and meteorological data are derived from the CAMELS-US dataset[49]. The dataset is accessible via: https://doi.org/10.1016/0022-1694(89)90184-4.

The source code and processed datasets are available in the GitHub repository: https://github.com/Polyethylenekjc/Sparse-Transformer-In-Streamflow-Prediction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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